

TAMIL NADU STATE BOARD - STANDARD 12 MATHEMATICS VOLUME 1

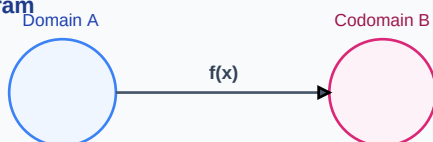
CHAPTER 3: THEORY OF EQUATIONS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 12 English Medium
Subject:	Mathematics Volume 1
Chapter Name:	Theory of Equations

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Polynomial Equation	An algebraic equation of the form $a_n \cdot x^n + \dots + a_1 \cdot x + a_0 = 0$, where coefficients are real or complex.
Rational Root Theorem	If a rational number p/q in lowest terms is a root of a polynomial with integer coefficients, then p divides a_0 and q divides a_n .
Descartes Rule of Signs	A technique to determine the maximum number of positive and negative real roots of a polynomial function.

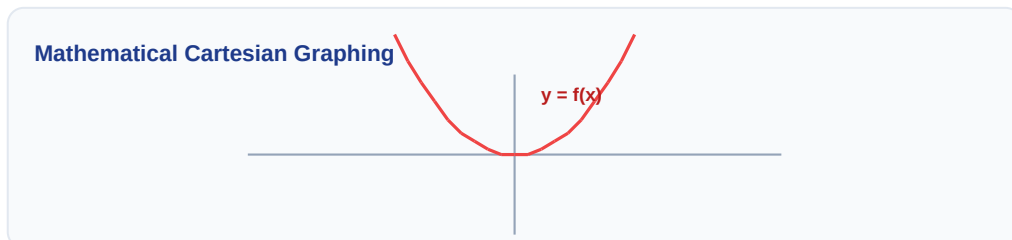
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Vieta's (Quadratic): $\alpha + \beta = -b/a$ | $\alpha * \beta = c/a$
- Vieta's (Cubic): $\sum a_i = -b/a$ | $\sum a_i * a_j = c/a$ | $a_1 * a_2 * a_3 = -d/a$
- Complex Conjugate Roots: If $a + ib$ is a root, then $a - ib$ is also a root (for real coefficients)

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Solve $x^3 - 3x^2 - 10x + 24 = 0$ given that one root is 2.

Step-by-step Solution:

Divide by $(x - 2)$ using synthetic division.

The quotient is $x^2 - x - 12 = (x - 4)(x + 3)$.

Therefore, the roots are $x = 2, 4, -3$.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Form a polynomial equation of lowest degree with rational coefficients having $2 + \sqrt{3}$ and $3 - 2i$ as roots.

Analytical Solution Strategy:

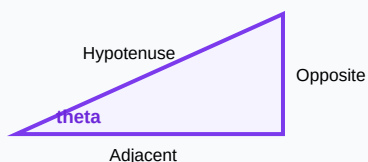
Since coefficients are rational, conjugate roots must occur: $2 - \sqrt{3}$ and $3 + 2i$.

Factors for quadratic 1: $(x - (2 + \sqrt{3}))(x - (2 - \sqrt{3})) = x^2 - 4x + 1$.

Factors for quadratic 2: $(x - (3 - 2i))(x - (3 + 2i)) = x^2 - 6x + 13$.

Polynomial is $(x^2 - 4x + 1)(x^2 - 6x + 13) = x^4 - 10x^3 + 38x^2 - 46x + 13 = 0$.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): If α and β are roots of $x^2 - 5x + 6 = 0$, find the value of $1/\alpha + 1/\beta$.

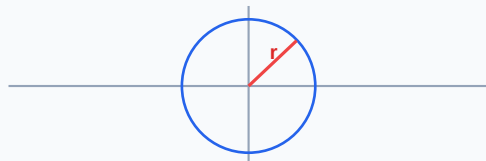
Q2 (2-Mark): Use Descartes' rule to analyze roots of $x^5 - 2x^2 + 5 = 0$.

Q3 (2-Mark): Find the sum and product of roots for $3x^3 - 5x^2 + 2x - 7 = 0$.

Q4 (5-Mark): Form a quadratic equation with roots $3 + i$.

Q5 (5-Mark): Solve the equation: $2x^3 - 9x^2 + 10x - 3 = 0$.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Control Engineering

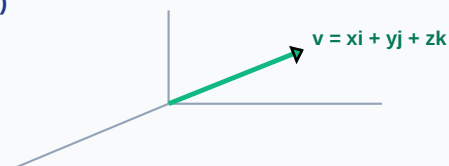
Polynomial roots evaluate pole placements, predicting mechanical control loop stability limits.

Application Case: Acoustic resonance

Theory of equations maps natural harmonic peak coordinates in audio chambers.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

